6061 Introduction to Deep Learning

Exercise 4

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This exercise sheet will not be graded and is provided for practice only.

1. **Computation graphs and backpropagation.**

   This exercise is based on the Deep Learning course from the University of Tübingen.

   For the following three expressions, draw the computation graph and apply the backpropagation algorithm to calculate the gradients of all nodes. For drawing the computation graph, each node should introduce a new intermediate variable and represent a single operation. Nodes should be reused wherever possible. For backpropagation, you do not need to simplify or analytically calculate the derivatives; it suffices to write down the expressions for the gradients.

(a) $\mathcal{A} = \frac{e^a}{\sqrt{b}} \log(a + b)$

(b) $\mathcal{B} = \sqrt{a+b+c^2} + \log(a+b+c^2) + \frac{a+b+c^2}{bc^2}$

(c) $\mathcal{C} = \tanh(\mathbf{W}\mathbf{x} + \mathbf{b}) \odot \sigma(\mathbf{M}\mathbf{x} + \mathbf{d})$. tanh and σ are applied element-wise, $\mathbf{W}, \mathbf{M} \in \mathbb{R}^{m \times n}, \mathbf{x} \in \mathbb{R}^n, \mathbf{b}, \mathbf{d} \in \mathbb{R}^m$, and $\odot$ is the element-wise multiplication. You can use matrix and vector operations as single nodes.

2. **Implement automatic differentiation.** Fill in the Juypter notebook 04_hw_tasks.ipynb to implement backpropagation for simple cases (no vectors) yourself.